

Scaling Behaviors for the Number of Bankrupted Firms

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The scaling properties of the probability distribution function for the asset and debts of the bankrupted firms and banks have become increasingly interesting over time[1-4]. We study on the probability distribution function of the number of bankrupted firms in Korea and USA. In Korea we consider the bankrupted firms in the period from 1 august 2002 to 28 october 2003. We also consider the bankrupted firms (chapter 11 and chapter 7) in the period 1 july 1986 to 29 january 2007 in USA. We calculated the cumulative probability distribution function(PDF) for the number of bankrupted firms. We observed the power law of the cumulative probability distribution function such as $P(x > n) \sim n^{-\alpha}$ with $\alpha = 0.91(2)$ in Korea and $\alpha = 8.9(1)$ in USA where n is the number of bankrupted firms.

In USA we also consider the asset and the number of employee in the bankrupted firms per day. We observe that the cumulative PDF of the asset and employee follows power law. In USA we rescale the time series by the inflation rate. However, the cumulative PDF of the asset and employee do not change the exponents of the power law by the rescaling of the time series.

Keywords

power law, bankruptcy, distribution function

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